

Markscheme

Mathematics: analysis and approaches
Practice question bank

112. (a)

Consider the expansion of $\sqrt{4-x}$ in ascending powers of x .

Find the coefficient of x^2 in this expansion.

write $\sqrt{4-x} = 2 \left(1 - \frac{x}{4}\right)^{\frac{1}{2}}$ (M1)

expand: $2 \left[1 - \frac{1}{8}x - \frac{1}{128}x^2 + \dots\right]$ (M1) A1

coefficient of x^2 is $2 \times \left(-\frac{1}{128}\right)$ (M1)

$= -0.015625$ A1

[5 marks]

(b)

Hence state the range of values of x , in the form $|x|$

validity requires $\left|\frac{x}{4}\right| < 1$ (M1)

$|x| < 4$ A1

[2 marks]